

MDS-UPDRS Finger Tapping Analysis

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [125, 123, 116, 127, 128, 137, 120, 127, 114, 143]

Tap amplitudes (pixels): [145.64683312726027, 149.4958193395387, 140.06427096158393, 140.34600101178515, 138.3979768638256, 136.24243098242192, 144.03124660989366, 136.60527076214885, 138.80922159568507, 136.40014662748717, 138.45215780189199]

GPT-5

MDS-UPDRS Item 3.4 – Finger Tapping Scoring Analysis

What the features show

- Trials expected: 10 taps; detected taps: 11. This often happens due to detection granularity (an extra small peak). We focus on the first 10 full inter-tap periods and the amplitude trend across the sequence.
- Inter-tap periods (frames): [125, 123, 116, 127, 128, 137, 120, 127, 114, 143]
 - Mean $\approx$ 126 frames; no clear progressive lengthening.
 - Variability is modest (min 114, max 143); there is one slower interval at the end (143), but otherwise periods fluctuate without a marked trend.
 - No evidence of multiple clear “interruptions” or a distinct arrest/freeze (there are no outliers suggesting a pause far beyond others).
- Amplitudes (pixels) across 11 taps:
 - [145.6, 149.5, 140.1, 140.3, 138.4, 136.2, 144.0, 136.6, 138.8, 136.4, 138.5]
 - Pattern: After a relatively larger peak at tap 2, amplitudes reduce modestly by tap 6, rebound at tap 7, then remain in a similar range. This is a mild fluctuation rather than a monotonic decrement from early taps. There is no clear, sustained decrement beginning right after the first tap, and no consistent mid-sequence progressive decrement; instead, it's small up-and-down variation around a slightly lower plateau after tap 2.

Mapping to scoring criteria

- Interruptions/hesitations:
 - No clear evidence of 3–5 interruptions or >5 interruptions; periods do not suggest repeated hesitations or a freeze.
- Speed:
 - Only slight variability; does not indicate moderate slowing.
- Amplitude behavior:
 - No consistent decrement beginning after the 1st tap (would support score 3).
 - Not a clear decrement starting mid-sequence that persists without rebound (required to justify score 2c).
 - Slight reduction near the latter half with small rebound suggests, at most, slight decrement toward the end.

Given these observations, the performance is near-normal with at most slight irregularity: small end-sequence slowing on the last interval and minor amplitude reduction without sustained decrement.

Score: 1
